

Progress Report

2026.01.06

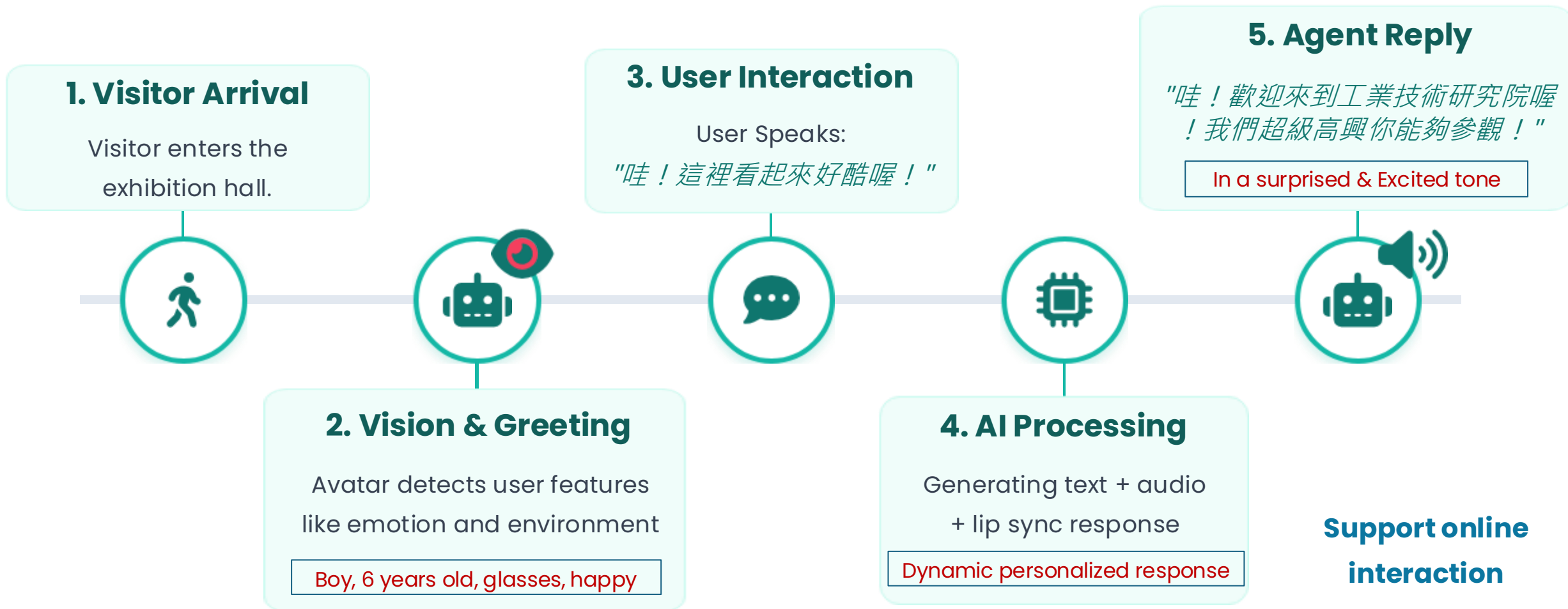


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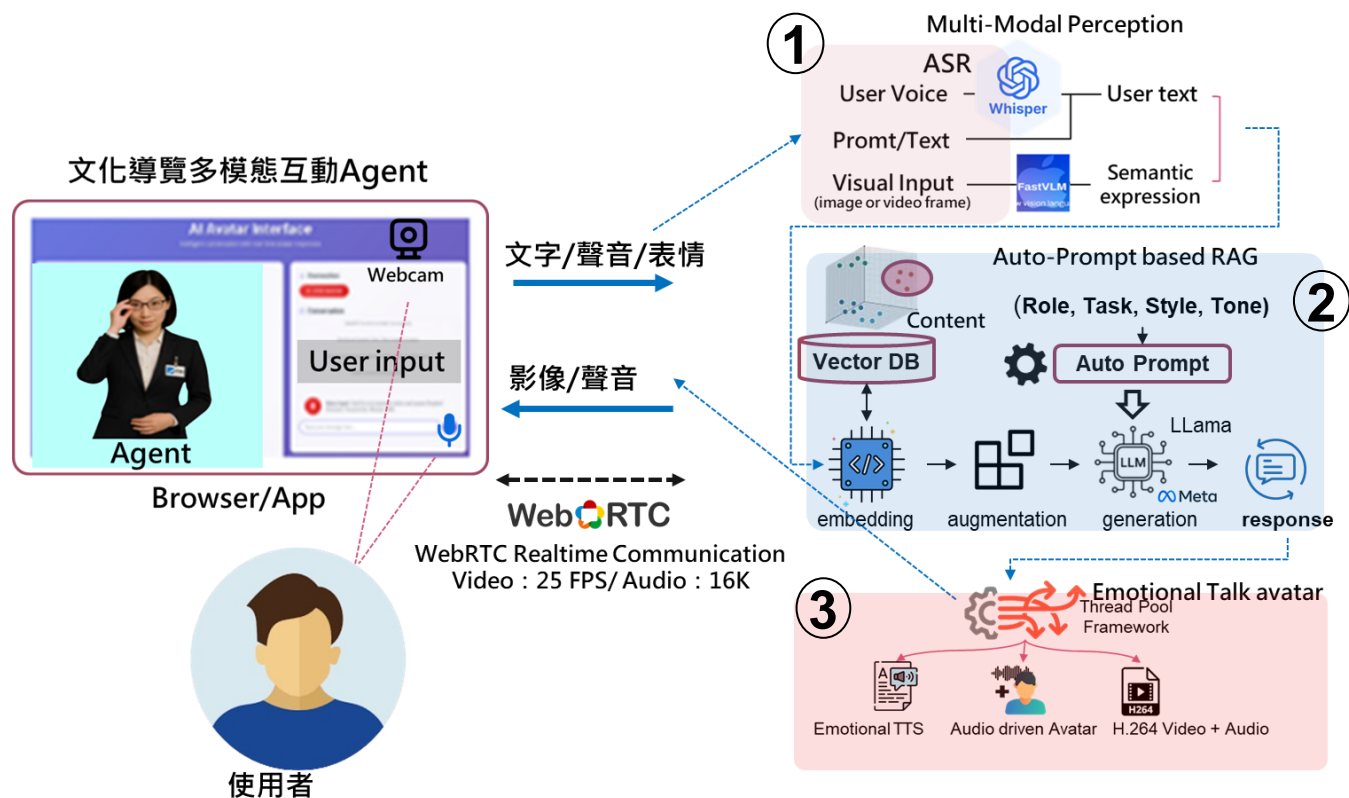
Project's problem setting

- Cultural Museum AI Guide Scenario



Project's system architecture

- LLM module focus on step ②

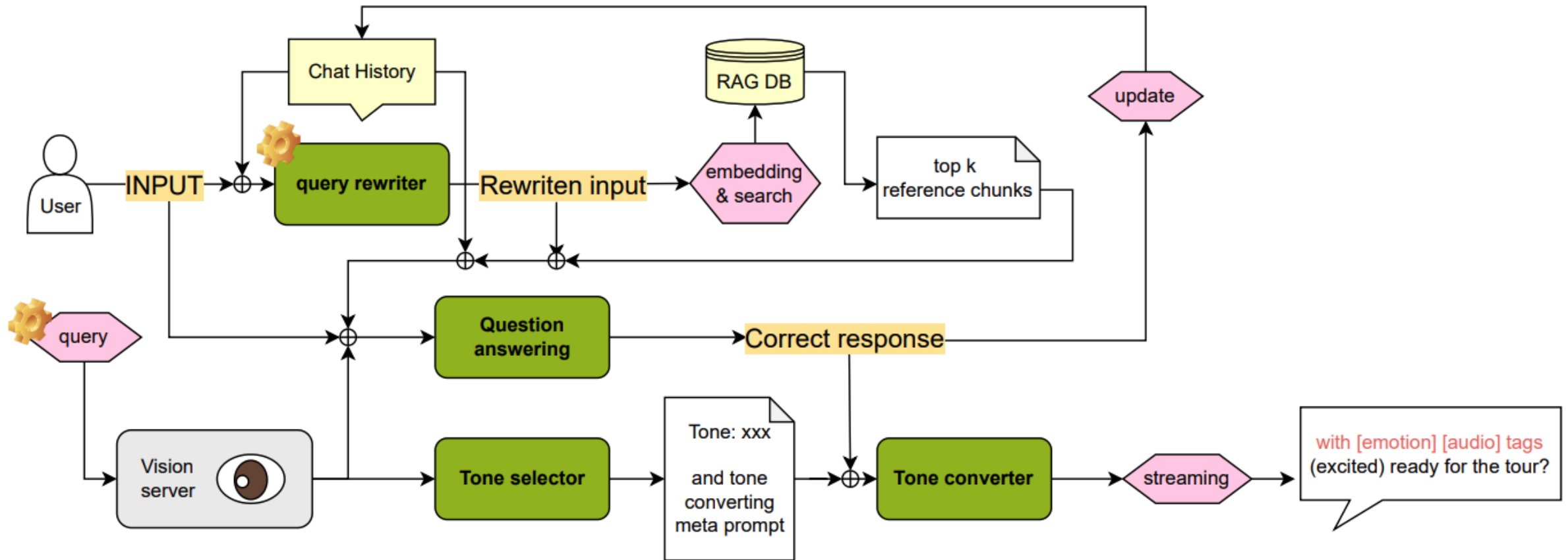


Feature to include in LLM module:

1. Fetch user description from vision server
2. A background knowledge database
3. User query rewriting agent (for embedding)
4. Question-Answering agent (correct answer)
5. **Tone selector**
6. Tone convertor

[Note]: auto prompt is not yet included;
currently we select a fixed tone converting
prompt based on the user description

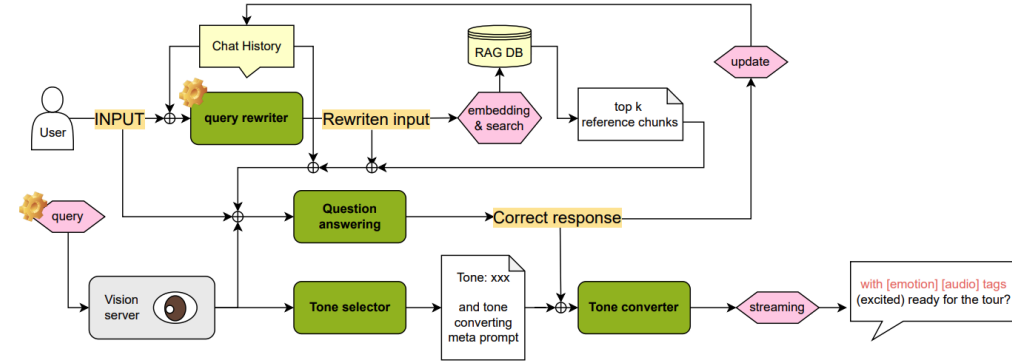
LLM module pipeline



LLM module – RAG support

- RAG Database
 - ChromaDB, each chunk in json format
 - Data sources
 - Official webpage: <https://www.itri.org.tw/index.aspx>
 - Wiki pages: [工業技術研究院 - 維基百科，自由的百科全書](#)
 - Virtual showroom: <https://www.itri-showroom.com/>
- Embedding model and method
 - Ollama embedding model: bge-m3:latest (*better Mandarin support!*)
 - Hybrid search [dense:sparse = 7:3]
 - Pick top k items
- Additional support – query rewriter

Rewrite the **follow-up** query into a complete question to enhance the rag support!

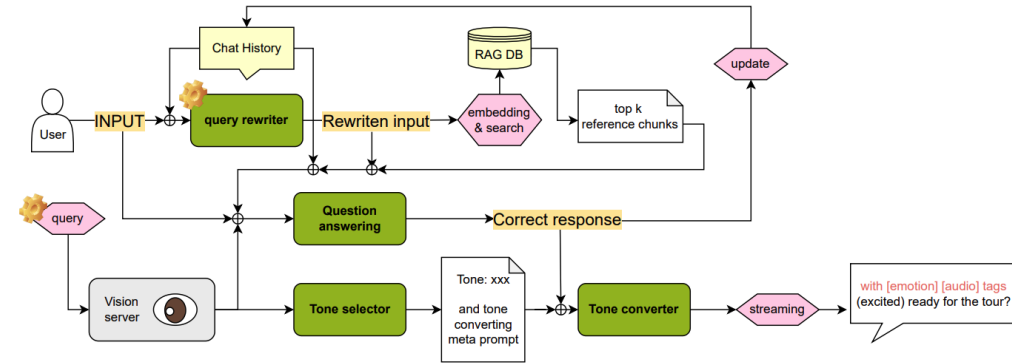


LLM module – QA response

- LLM response model
 - Ollama - llama3.1:70b-instruct-q4_0
- Format:
 - **Input:** *user input $\oplus$ rewritten input $\oplus$ chat history $\oplus$ reference data from database $\oplus$ user description*
 - **Output:** *a response with correct data*
- Prompt:

Role playing, target specify, limitation, input and output format, reasoning guidance, few shot

 - *in API/tone_system_prompt_no_tag.py*



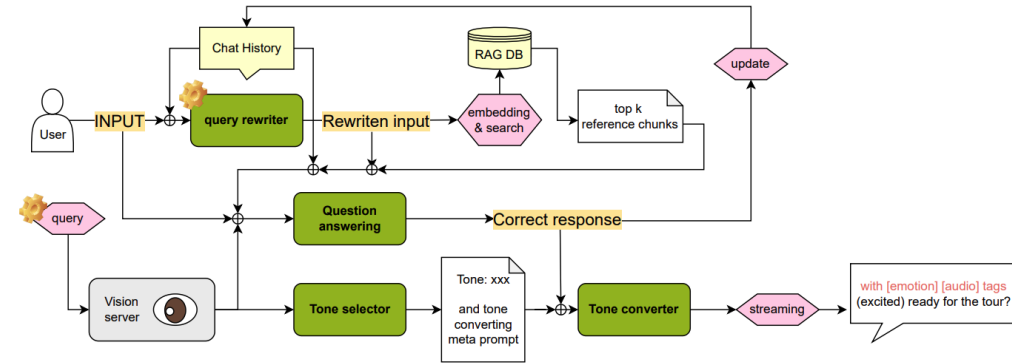
LLM module – Convert tone

• Tone selector

- Key decider: age, style
- Tone supported: *elder / kid / adult-professional / adult-casual*
- **Input:** *user_description*
- **Output:** *corresponding tone converting guidance (meta prompt)*
- *Note: not yet implement the auto-prompt method to dynamically construct prompt*

• Tone converter

- Based on the convert guidance's meta prompt, convert the “correct” response into a human-like, with emotion control, customized response.
- Use tts → response with audio tag in *FishAudio*; use voice cloning → no tag

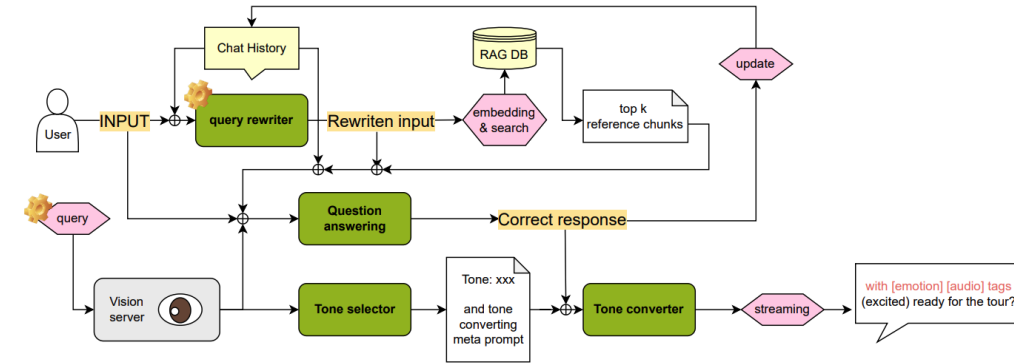


LLM module – Convert tone

- Prompt Format:

Role specified, **goal** clarified, target **language** used, tone converting **guidance** like what is storytelling and how to tell a story, **greeting** method, **few shot**, output format and restriction

- *in API/tone_system_prompt_no_tag.py*



****Example 1: Specific Technology (Connecting to previous context)****

– User Input: "你可以再解釋得清楚一點嗎?" (Refers to 生態樹)

– Part 1 Fact: "生態樹樹幹1利用6台風扇每小時循環空氣7.7次，降低室溫。"

– Cultural Output:

您說的對，其實這棵大樹背後有很深的情感與科學。

說起這棵樹的運作方式啊，您可以想像一下，在樹冠頂端藏著六台非常安靜的電風扇。它們不停地運作，每小時可以幫我們把室內的空氣循環將近八次之多。

這代表什麼呢？這代表即便展示區人很多，空氣依然能像在森林裡一樣新鮮，而且能把夏天的悶熱都吹散。我們工研院研發這項技術，就是希望能讓大家在室內也能呼吸到像山間清晨一樣涼爽的空气。

LLM module – current issue

- LLM reasoning dilemma
 - Use **ollama server** currently → can run better reasoning module with no additional output problem, but **slow start**.
 - Use **vllm** → **faster response time** to support realtime feature, but have to run smaller model due to hardware
- Auto prompt methodology
 - Use quick select right now. (a simple function $f(\text{user description})$)
 - If (elder) return elder_tone_converting_system_prompt
 - If (kid) return kid_tone_converting_system_prompt
 - Need to switch to auto prompt to support various kind of user
 - May have more latency for constructing the tone converting's meta prompt
 - Model being used should have good understanding to provide suitable converting guidance